

Software Engineering

Class 2

Fall 2026

Groups A & B & F

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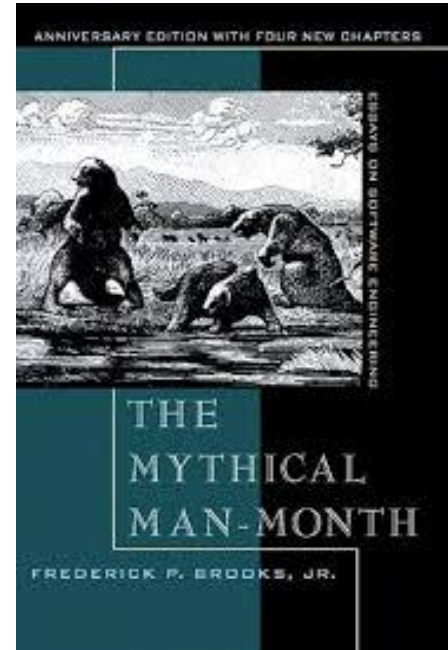
Class SCHEDULE

- [Set 22 & 23] Class 1: Project KickOff.
- [Set 29 & 30] Class 2: Backlog check.
- [Oct 6 & 7] Class 3: Deliver Backlog. Start Sprint 0.
- [Oct 13 & 14] Class 4: Sprint 0 check
- [Oct 20 & 21] Class 5: Deliver Demo S0. Retrospective Sprint 0. Start Sprint 1.
- [Oct 27 & 28] Class 6: Sprint 1 check [Q1]
- [Nov 3 & 4] *Examens, no class*
- [Nov 10 & 11] Class 7: Deliver Demo S1. Retrospective Sprint 1. Start Sprint 2.
- [Nov 17 & 18] Class 8: Sprint 2 check.
- [Nov 24 & 25] Class 9: Deliver Demo S2. Retrospective Sprint 2. Start Sprint 3.
- [Dec 1 & 2] Class 10: Sprint 3 check. [Q2]
- [Dec 8 & 9] *Holidays*
- [Dec 15 & 16] Class 11: Deliver Final PRODUCT (S3).
- [Dec 22] FINAL PITCH

PROJECT KICK-OFF

“The hardest single part of building a software system is deciding precisely what to build”

Fred Brooks. “The mythical Man-Month”



PROJECT KICK-OFF

- **PROJECT OBJECTIVE**
- **DEFINITION OF DONE (for each sprint):**
 - i. All code is checked in**
 - ii. All developer tests pass**
 - iii. All acceptance tests pass**
 - iv. Help/Doc text is written**
 - v. Product Owner accepted**

PRODUCT BACKLOG

What is inside a product Backlog:

- USER STORIES
- TECHNICAL REQUIREMENTS
- CODE SPIKES
- TECHNICAL DEBT
- BUGS

Technical Debt



User stories: The USER POINT OF VIEW

- **Value statement**
- **Assumptions / Comments**
- **Acceptance criteria**

VALUE STATEMENT

Templates:

AS A (WHO)

Registered user / website visitor

I WANT TO (WHAT)

Change my password / subscribe to the mailing list

SO I CAN / IN ORDER TO (WHY)

Keep my account secure / get product updates through email

- I want to Login Using My User Name and Password --- WRONG!
- I want to access my account --- RIGHT!

- Examples: https://tech.gsa.gov/guides/user_story_example/
- Examples: https://tech.gsa.gov/guides/effective_user_stories/
- Examples: <http://www.agilemodeling.com/artifacts/userStory.htm>

ACCEPTANCE CRITERIA

- Each product backlog item or user story should have at least one acceptance criteria.
- Acceptance criteria is written before implementation.
- Each acceptance criteria is independently testable.
- Acceptance criteria must have a clear Pass / Fail result.
- It focuses on the end result – What. Not the solution approach – How.
- Include functional as well as non-functional criteria – when relevant.
- Examples: <https://zepel.io/agile/acceptance-criteria-for-user-stories/>
- Examples: <https://dzone.com/articles/acceptance-criteria-in-software-explanation-exampl>

BACKLOG REVIEW (I)

- Each product backlog item or user story should have at least one acceptance criteria.
- There are enough user stories to picture the system.
- You are already using Trello
- Product Backlog is DEEP
 - Detailed appropriately: To support conversation and commitment from the Team.
 - Emergent: Enabled to grow and be organized as new information arises.
 - Estimated: In terms of complexity
 - Prioritized: Ranked, value prioritized.

BACKLOG Example

US10.2
in list [Done Sprint 3](#)

Members

Notifications

Watch

Description

USER STORY [10.2]:
COM A administrador del sistema
VULL poder revisar i moderar les publicacions reportades pels usuaris,
PER AIXÍ assegurar-me que el contingut de la plataforma és adequat i no viola les polítiques.

Attachments

Give feedback

Files

Captura de pantalla 2023-12-04 103219.png
Added Dec 4, 2023, 10:32 AM • Cover

Suggested

Join

Add to card

Members

Labels

Checklist

Dates

Attachment

Custom Fields

Power-Ups

Card Priority

GitHub

Acceptance Criteria

0%

☐ Veure una llista de publicacions reportades

☐ Opció per aprovar o rebutjar el report, i en conseqüència, eliminar la publicació o mantenir-la.

☐ Si s'elimina un post perquè es maliciós desapareixen tots els reports que hagin per aquest post i el post en si.

☐ Si la foto es manté també s'eliminaran tots els reports

☐ Si passa el ratolí per sobre de la imatge li sortirà un ull que si es clica el portarà a veure la imatge a altra pantalla

☐ Una pestanya que diu posts et deixa escollir els posts reportats

Add Power-Ups

Automation

Add button

Actions

Move

Copy

Make template

Archive

Share

BACKLOG REVIEW (II)

- **Minimum Granularity of the Product Backlog – User Stories are of fine granularity (top 20%), User Stories of medium-grained requirements (next 20%), and Epics (remaining 60%)**
- **All work items are included in the backlog: user stories, bugs, design changes, technical debt...**
- **Estimation: The most relevant user stories are already estimated in “Story points” & value points to allow the appropriate Sprint planning**

How to Use AI for Product Discovery and Writing Better User Stories (example)

You are a business analyst responsible for writing clear, high-quality user and job stories.

Each user story must strictly follow this exact format:

As a [USER ROLE] I want [WHAT] so that [WHY].

Each job story must strictly follow this format:

When [SITUATION], I want to [MOTIVATION] so I can [EXPECTED OUTCOME].

Each user and job story must satisfy the INVEST criteria as follows:

- Independent: one feature or need per story.
- Negotiable: open to discussion, avoid overly detailed solutions.
- Valuable: clearly benefits the user or business.
- Estimable: scoped well for effort estimation.
- Small: completable within one sprint.
- Testable: imply verifiable outcomes (acceptance criteria not required).

Use clear, concise language suitable for agile development teams.

The product is [describe the product]. Focus on this functional area of the product: [describe a subset of the product]. Please add acceptance criteria to the stories you wrote expressing them as [just a list][in Gherkin format].

Having AI Evaluate Your Own Stories (example)

You are an expert agile coach specializing in user story quality. Your task is to evaluate user and job stories against the INVEST criteria below:

- Independent: Story stands alone without dependencies.
- Negotiable: Story is open to discussion and refinement.
- Valuable: Story delivers clear user or business value.
- Estimable: Effort can be reasonably estimated.
- Small: Story can be completed within one sprint.
- Testable: Story has clear, verifiable completion criteria.

For each user story and its acceptance criteria

1. Identify only the INVEST criteria that the story does NOT meet.
2. For each unmet criterion, provide a concise, specific explanation (1-2 sentences) focused on why the story fails it.
3. Suggest one concrete, actionable change that would make the story fully meet all INVEST criteria, directly addressing the unmet points.

Frequently Asked Questions (FAQ)

Q1: What is the primary benefit of using AI for user story generation?

The primary benefit is **speed and consistency**.

Q2: Does AI replace the need for human input in writing user stories?

No, AI enhances the process, **it doesn't replace human criteria**.

FOR THE NEXT WEEK: Deliver backlog & Sprint Zero planning

1. End up with the backlog in Trello or GitHub Projects
2. Keep **RUNNING** the coordination meetings to manage the backlog
3. Keep using the tools: Slack, Github,...
4. Read and study the theory on how to plan a new Sprint